More Experienced Intermediaries - More Principal-Agent Problems?*

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Abstract

Both the seller (principal) and the real estate agent (agent) would like a transaction to happen at the highest possible price. However, in the case of a failed sale, a seller continues to own the property and can attempt to sell it again, while an agent often loses the client and walks away with a bad record. Consequently, agents care more about closing the transaction and are likely to incentivize sellers to lower their asking price to attract more buyers. Using proprietary data on for-sale listings in the U.S., I find that the principal-agent problem is more severe among experienced agents. However, experienced agents are more successful at selling homes for their clients at any price point. Informed by these findings, I build a housing search model to formalize the trade-off between prices and sale probability. I allow agents' matching ability to vary by experience via differential "matching technology". A seller working with an experienced agent would prefer to take advantage of this technology premium and list their home at a particularly high price. However, experienced agents especially dislike the no-sale outcome because of reputation concerns and their opportunity cost of time. As a result, they choose to list the homes at much lower prices relative to sellers' optimum. Despite that, sellers are still better off with top agents. They are indifferent between paying 3% to a new intermediary and as much as 12% to a top agent. In the final exercise, I find that an alternative commission structure with stepwise-linear commissions is able to close 76% of the welfare gap caused by the principal-agent problems with top agents.

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1 Introduction

The principal-agent problem is common in industries where agents are paid conditional on transaction. Agents are incentivized to close deals and do so quickly, which may not align with the principal's goal of finding the best match or securing the best terms. Several studies have addressed the principal-agent problem in the financial advising industry (Linnainmaa et al. (2015),Linnainmaa, Melzer, and Previtero (2021)) and in real estate intermediation (Levitt and Syverson (2008); Rutherford, Springer, and Yavas (2005), Rutherford, Springer, and Yavas (2007), Xie (2018)).[1] Seminal papers on the topic rely on comparing strategies that agents employ for themselves and those that they take on for their clients. This exercise omits an important consideration - the agent's experience. In this paper I examine how the extent of the principal-agent problem varies with the agent experience in real estate intermediation and how the misaligned incentives contribute to the overall value or real estate listing agents.

Real estate agents in the U.S. have long been subject to criticism over agency problems. In light of the recent NAR settlement in 2024 and the preceding court cases, the compensation structure of real estate agents has once again come under scrutiny from the media and policy makers. Critics say that the service agents provide does not justify the elevated commissions they charge, in part due to misaligned incentives with their clients (the principal-agent problem). Agents are paid on commission, which is a fixed percentage of the sale price. As such, it is in their interest to secure the highest price, but more importantly, to make the deal happen at all. Although these goals generally align with the seller's objectives, the trade-off between price and sale probability might be different for the two parties and additionally vary within agents by experience. For example, an agent with an established reputation might be particularly reluctant to list at a high price risking a no-sale outcome. This could be either due to concerns about their reputation, or because they have many clients to work with and their opportunity cost of time is particularly high. On the other hand, experienced agents are potentially more efficient at their job and can provide better service to their clients. In this paper, I evaluate these trade-offs and derive a “fair" competitive commission rate for agents of different experience levels. I also propose and "optimal piecewise" commission structure that maximizes seller's welfare, taking the principal-agent problem into account.

This project has both an empirical and a theoretical parts. In the empirical part, I document the

[1] Additional industries with similar settings include recruiting agents, travel agents, insurance and mortgage brokers.

list pricing strategies and sales outcomes of listings represented by real estate agents of different experience levels. I use a proprietary dataset on for-sale home listings coming from thirty eight different Multiple Listings Platforms (MLSs). This data includes the universe of listings within the geographical area covered by each MLS from 2000 to 2020. For each observation (listing) there is very detailed information on the property for sale and the identifier for the listing agent. Since I observe the entire activity history of each real estate agent, I can construct a measure of their experience level and study how experience impacts the pricing decision and the probability of sale. To address selection concerns, I make use of a quasi random-assignment strategy used in Gilbukh and Goldsmith-Pinkham (2024). This strategy exploits the fact that sellers tend to use the same agent that helped them with purchasing the property. However, that buying agent might not be available at the time of the sale decision if they had left the profession. In that case, the seller tends to hire an average agent from the industry. As such, sellers whose original buying agent was very experienced but ended up exiting the market receive a negative shock to experience of their agent relative to a seller who had a similar buying experience and is able to go back to their experienced agent. On the contrary, sellers whose buying agent was not very experienced receive a positive shock when choosing a new agent from the pool relative to a seller whose inexperienced agent is still available for work. The IV strategy confirms the OLS results that more experienced agents are more likely to sell the property at every price point. However, they do not seem to take advantage of this experience benefit and tend to sell homes at same prices as inexperienced agents.[2] To quantify the principal-agent problem and examine alternative commission structures that might benefit sellers, I turn to a theoretical model of housing search.

In the theoretical part of the paper, I set up a directed housing search model of the housing market. In this model there are sellers, buyers, and listing agents. Sellers own a house but derive no value from it. They hire a listing agent and fully delegate the selling process of the property to them (including the choice of the list price). Although there is no heterogeneity in buyers or sellers, listing agents differ in their level of experience. The model follows a directed search framework where each listings belongs to a different sub-market indexed by the list price and list agent experience. Buyers can direct their search to these sub-markets, trading off the attractiveness of the price, the matching technology, and the potential competition from other buyers. All else equal, markets with higher posted prices will attract fewer buyers, which will result in a higher probability of purchase for those few buyers

[2] These results are consistent with Gilbukh and Goldsmith-Pinkham (2024), where the data sample is from 2000 to 2013 only.

who choose to participate in them. In contrast, markets with low prices will attract many buyers, which will reduce the chance of purchasing for each of them due to more competition. In addition to the prices, each market is also defined by the experience of the list agent. I allow for a flexible set up where experienced agents potentially have a more efficient matching technology, with more matches being made for the same number of buyers and sellers in the sub-market. I abstract from buyer agents in this model, but require that sellers pay a 3% commission to the buyer side as is the standard during the period of the empirical study.[3] I also abstract from the matching process between sellers and agents.[4]

Given the buyers' trade-off between price and sale probability, I set up the optimal list price problem from the perspective of the seller, conditional on the experience of their listing agent (i.e. conditional on the matching technology). Sellers working with experienced agents would like to exploit the superior matching technology and list their homes at particularly high prices. However, I assume that it is the agent who advises and ultimately decides which price the house will be listed at. Agents have an optimal pricing problem to maximize their commission from the sale. I allow agents of different experience level to vary in their dis-utility of the no-sale outcome, which is naturally the case because of either differential reputation concerns or due to the varying opportunity cost of time. Higher discrepancy between the seller- and the agent-optimal prices indicates a more severe principal-agent problem.

The model parameters are estimated to match the relative list prices and sale probabilities for each level of agent experience in the data. For every experience level, I then quantify the welfare loss for the seller arising from following the agent's list price advice rather than using seller-optimal pricing. This loss defines the extent of the principal-agent problem as a function of experience. I also estimate the overall value of working with listing agents of different experience. The latter incorporates the fact that experienced agents are better at matching their clients at any price, so their overall value will be a function of both the pricing decision and the matching efficiency.

Finally, the model allows me to examine different commission structures. First, I derive "fair" relative commission rates for each level of experience. I find that the best agents improve the welfare of the seller by up to 12% of the value of the property compared to agents with no experience. Since

[3] In other words, buyers are the ones directing their search towards different sub-markets based on their expected payoffs. An alternative interpretation is that buyers are represented by buyer agents who act in their clients best interest.

[4] My results show that working with a more experienced agent is clearly more beneficial for sellers. However, in the data, we see sellers working with agents of different experience levels. Gilbukh and Goldsmith-Pinkham (2024), for example, assumes that a fraction of clients matches with a random agent, while the complimentary fraction seeks a referral, meaning they match with an agent with a probability proportional to their experience share.

the commission rates in the U.S. have been notoriously fixed, [5], an alternative interpretation of this result is that sellers who work with inexperienced agents lose on average 9%[6] of the property value due to sub-par outcomes in the search market.

In the second commission exercise, I construct an optimal piecewise-linear commission function where marginal commission is a function of the sale price.[7] I use an iterative procedure assuming that agents receive $\psi_1 = 3\%$ up to p_1 which is the price that agent of experience one charges in the baseline model. For each subsequent level of experience $e \in \{2, 3, ..., 50\}$ I find the optimal ψ_e for the next price segment $p > p_{e-1}$ that maximizes the seller's welfare conditional on working with an agent of experience e. The corresponding agent-optimal price p_e defines the next notch in the commission function. This process constructs a commission schedule $\{\psi_1, \psi_2, ..., \psi_{50}\}$ and price notches $p_1, p_2, ..., p_{50}\}$. I find that the optimal marginal commission is as high as 12% for most experienced agents. This commission structure also significantly closes the welfare gap between agent- and seller- optimal pricing. When working with top agents, sellers can close this principal-agent gap by as much as 76%.

This paper is closely related to Gilbukh and Goldsmith-Pinkham (2024) which focuses on the probability of sale outcome for agents of different experience levels throughout the housing cycle. The model in Gilbukh and Goldsmith-Pinkham (2024) analyzes the experience distribution of real estate agents arising from endogenous entry and exit decisions of real estate agents across different periods in an aggregate housing cycle. In this paper, I abstract from the endogenous formation of the distribution of experiences and aggregate fluctuations in the market. Instead, I am able to add the pricing decision into the housing search model which enriches the discussion of value that agents provide to their clients and allows me to examine the principal-agent problem between sellers and listing agents. In addition, the setup allows me to discuss the consequences of different commission structures.

This paper contributes to the literature on the role of real estate agents in alleviating housing search frictions (Barwick and Wong (2019),Hendel, Nevo, and Ortalo-Magné (2009)). More specifically, this paper addresses the principal-agent problems in this market (Levitt and Syverson (2008), Kim

[5]See Barwick, Pathak, and Wong (2017). Although the NAR settlement aims to make the commissions more competitive in the future, the effect of the policy implemented in July 2024 is quite uncertain and the sentiment in the industry is that the aftermath of the changes is still shaking up.

[6]That is they would pay an additional 9% of the property value to work with an experienced agent on top of the 3% they would pay for a new agent.

[7]The total commission is equal to an integral of the marginal commission as a function of the price: $\psi(p) = \psi_1 \min(p, p_1) + \sum_{e \in \{2,3,...50\}} \psi_e(min(p, p_e) - min(p, p_{e-1}))$.

(2024),Han and Hong (2016))[8]. However, to my knowledge, the heterogeneity of real estate agents when it comes to the principal agent problem has not been previously explored. And yet, it is an important aspect of the market, because a significant portion of the intermediaries in real estate have no experience.

Finally, this paper develops a way to compute an optimal piecewise linear commission compensation structure that takes into account agent heterogeneity. This structure can provide guidelines for agent compensation after the NAR lawsuit, as many experts believe that the market will become more competitive. It can also be a useful tool applied to different types of sales intermediaries, e.g. finance brokers and job recruiters.

The rest of the paper is structured as follows. Section 2 sets out the theoretical framework. Section 3 describes the data and the empirical strategy for causally estimating the probability of sale and prices as a function of the experience of the listing agent. Section 4 outlines the calibration exercise. Section 5 presents the results. Section 6 describes the "fair" commission and optimal piecewise-linear commission calculations. Section 7 includes a brief discussion and Section 8 concludes.

2 Model

2.1 Model Setup

There are three types of agents in the model: buyers, seller, and real estate intermediaries (I will call them agents from now on). Buyers pay an entry fee c_b to participate in the housing market. If they successfully match and purchase a home, they derive value v from the purchase. I assume that if buyers are unable to match, their reservation value is zero. Each seller owns one house that they are hoping to sell. Beyond the expected sale proceeds, they derive no additional utility from their ownership of the house. To sell a house, a seller has to employ a real estate agent[9]. Agents differ by experience level e. Agent's experience determines their matching technology - more experienced agents can match their clients with higher probability, all else equal. In addition, agents set a price for the house with full commitment to selling at this price if the buyer is found. The housing market is characterized by sub-markets indexed by j, each differing in technology (experience e_j) and price, p_j. I adopt a directed search framework where buyers direct their search towards a particular sub-market

[8]Notable other applications with a similar setting are financial brokers as in Egan, Ge, and Tang (2022). Robles-Garcia (2019) examines principal-agent problems among mortgage brokers.

[9]Buyers may also delegate their purchase to an agent, but that doesn't change the result as the heterogeneity of agents only affects the seller side

j[10].

In each submarket, j with s seller agents and b buyers, $s(1-e^{-b\nu(e_j)/s})$ matches are realized, where e_j is experience level of listing agents in that market and p_j is the price posted in this market.[11] The function $\nu(e)$ captures the experience advantage of listing agents. Power functions are useful in this setting, as they allow for a decreasing returns to scale, meaning faster "learning" by inexperienced agents observed in the data.[12]

The match probability for a buyer and a seller in market j is a function of listing agents experience e_j and the market tightness, $\theta_j = b/s$:

$$\eta(e,\theta;p) = \frac{1}{\theta}\left(1-e^{-\nu(e)\theta}\right) \qquad \text{Buyer Match Probability} \qquad (1)$$

$$\mu(e,\theta;p) = 1-e^{-\nu(e)\theta} = \theta\eta(e,\theta) \qquad \text{Seller Match Probability} \qquad (2)$$

Note that while p_j doesn't enter the match probability functions, it endogenously affects them through market tightness. A higher posted price will deter buyers from entering the sub-market and will lower the tightness, making matches less likely for the seller. Formally, each buyer chooses a sub-market to maximize buyer valuation:

$$V^B = -c_b + \max_j \eta(e_j,\theta_j;p_j)(v-p_j). \qquad (3)$$

Recall that c_b is the entry cost and v is the buyer's valuation of the property. In equilibrium, participating in each active sub-market will deliver the same ex ante utility to buyers. If that was not the case, then buyers would only enter the highest utility sub-markets, rendering others inactive. Intuitively, while some markets have better technology or more favorable prices, they also attract longer lines, equalizing the overall outcome for each buyer. The free entry condition implies that the buyers will choose to participate (and crowd each other out) until there are enough buyers that $V^B = 0$. Together with the equilibrium condition of equal surplus, this free entry condition determines the trade-off between technology, queue, and prices:

[10]While our model's setup and solution method echoes the standard directed search model (see Moen (1997) and Shimer (1996)), it differs in a significant way. The standard directed search model involves buyers sorting on price only (or wage in the labor literature), while here, I allow for buyers to sort on both price and matching efficiency.

[11]This matching function is an approximation of an urn-and-ball matching function for a large number of buyers and sellers. The probability of match is restricted to be between zero and one. In this formulation match probabilities for both buyers and sellers exhibit constant return to scale, meaning that Ican express them as a function of the ratio of buyers to sellers. For a more detailed discussion, refer to Rogerson, Shimer, and Wright (2005).

[12]Other papers that use power functions to describe experience effect on production include Benkard (2000), Kellogg (2011), and Levitt, List, and Syverson (2013).

$$p_j = v - \frac{c_b \theta_j}{1 - e^{-\nu(e_j)\theta_j}} \tag{4}$$

I next turn to the optimal pricing decision for each experience level of the listing agent. I first solve for an optimal price from the seller's perspective. I then turn to the agent's problem and solve for a price that they ultimately set for their clients.

The two prices are different, because of differences in value of a "failed sale". Sellers have a continuation value of holding on to the property that failed to sell. Agents, on the other hand, have a cost associated with a no-sale outcome. I can interpret this cost in one of two ways. First, I can think of this cost as an opportunity cost of time - not only the agent misses out on the commission of the property does not sell, they also have lost valuable time that they could have spent serving another client. Second, this cost could be viewed as a reputation cost. An agent wants to avoid having history of not being able to sell a property on their professional record, because that can cost them future clients. Additionally, if the seller client is not able to sell their house, they are less likely to recommend the agent or write a good review. In either interpretation, it is reasonable to assume that the cost is higher for experienced agents.

2.2 Seller's Optimal Price

If a seller who is matched with an agent of experience e were setting a price for the listing, they would be solving the following optimization problem:

$$\begin{aligned} \max_{\theta,p} \quad & \mu(\theta,e)(1-\psi-3\%)p + (1-\mu(\theta,e))\beta V^S_{agent_p}(e) \\ & s.t. \\ & p = v - \frac{c_b\theta}{1-e^{-\nu(e)\theta}} \end{aligned} \tag{5}$$

Conditional on listing agent experience e and given a particular price and market tightness θ, a seller successfully sells their home with probability $\mu(\theta, e)$ and gets the price p net of commission ψ to the seller agent, and 3% to a buyer agent.[13] With complimentary probability, they get the discounted continuation value $V^S_{agent_p}(e)$ of staying a seller in the following period.[14] A seller is maximizing this

[13]While buyer agents are not explicitly modeled here, this part of the commission was standard in the industry during the period of my data.

[14]I assume that remaining a seller implies matching with an agent of the same experience. Alternatively, a seller could

payoff function over the price and the market tightness, taking the buyer trade-off between the price and the tightness for each technology level (4) as given.

Substituting μ from (2) and p from (4), the maximization problem becomes:

$$\max_{\theta} \left(1 - e^{-\nu(e)\theta}\right) v(1-\psi-3\%) - c_b\theta(1-\psi-3\%) + e^{-\nu(e)\theta}\beta V^S_{agent_p}(e) \tag{6}$$

The first order condition pins down a unique solution to the optimal tightness $\hat{\theta}^s_e$ in each sub-market:

$$\nu(e)e^{-\nu(e)\hat{\theta}^s_e}(v(1-\psi-3\%) - \beta V^S_{agent_p}(e)) = c_b(1-\psi-3\%) \tag{7}$$

$$\hat{\theta}^s_e = \frac{1}{\nu(e)} \log\left(\frac{v\nu(e)}{c_b} - \frac{\beta V^S_{agent_p}(e)\nu(e)}{c_b(1-\psi-3\%)}\right) \tag{8}$$

Note that the continuation value V_S of the seller puts a downward pressure on the $\hat{\theta}^s_e$ which means that the optimal prices for the agent will be such as to draw fewer buyers to the market, i.e. higher. Plugging in $\hat{\theta}^s_e$ in the price equation leads to:

$$\hat{p}^s_e = v - \frac{c_b \log\left(\frac{v\nu(e)}{c_b} - \frac{\beta V^S_{agent_p}(e)\nu(e)}{c_b(1-\psi-3\%)}\right)}{\nu(e) + \frac{c_b(1-\psi-3\%)}{v(1-\psi-3\%)-\beta V^S_{agent_p}(e)}} \tag{9}$$

In summary, for each experience level, the sub-markets collapse to a unique market with a particular tightness $\hat{\theta}^s_e$ and price $\hat{p}^s_e$. I can compute the derivative of $\hat{p}^s_e$ with respect to e as a product of two derivatives: $\frac{d\hat{p}^s_e}{de} = \frac{d\hat{p}^s_e}{d\hat{\theta}^s_e}\frac{d\hat{\theta}^s_e}{de}$. From equation (4),

$$\frac{d\hat{p}^s_e}{d\hat{\theta}^s_e} = -\frac{c_b(1 - e^{-\nu(e)\hat{\theta}^s_e}(1 - \hat{\theta}^s_e\nu(e))}{\left(1 - e^{-\nu(e)\hat{\theta}^s_e}\right)^2} < 0 \tag{10}$$

There is a negative relationship between the market tightness and the price, because higher prices deter buyer from entering the sub-market.[15]

draw a random agent from the distribution.

[15] Since $\nu(e)\hat{\theta}^s_e > 0$, then $e^{\nu(e)\hat{\theta}^s_e} > 1 - \nu(e)\hat{\theta}^s_e$ and $1 > e^{-\nu(e)\hat{\theta}^s_e}(1 - \hat{\theta}^s_e\nu(e))$. The numerator and the denominator are thus both positive and so the overall derivative is negative.

$$\frac{d\hat{\theta}^s_e}{de} = \frac{\nu'(e)}{\nu^2(e)} \left(1 - \log\left(\nu(e)\frac{v(1-\psi-3\%) - \beta V^S_{agent_p}(e)}{c_b(1-\psi-3\%)}\right)\right) \tag{11}$$

This derivative is also negative for reasonable parameter values.[16] Meaning that the total derivative of optimal price with respect to experience is positive. This means that sellers who are matched with more efficient agents would prefer to list the homes for higher prices.

The expected welfare of sellers matched with an experience e listing agent who lists at their desired price is:

$$V^S_{seller_p}(e) = \mu(\hat{\theta}^s_e, e)(1-\psi-3\%)\hat{p}^s_e + (1-\mu(\hat{\theta}^s_e, e))\beta V^S_{agent_p}(e)$$

I'll come back to defining $V^S_{agent_p}(e)$ in the following section.

2.3 Agent's Optimal Price

I have derived a price that a seller would optimally set if they had access to their agent's matching technology and were fully informed about market conditions. However, sellers rely on their agent's expertise and fully delegate the pricing decision. Both the seller and the agent desire for the sale to happen and, all else equal, prefer a higher price outcome. However, the two parties have a different valuation for a failed sale outcome. While sellers get to try their luck in the next period, agents might not get a chance to represent the client again in the future. I assume that $\alpha(e)$ is the probability that the seller re-hires the same agent if the property fails to sell. When calibrating the model, I will compute this probability as the fraction of re-list agents that have the same agent as the original listing. In addition to losing a client, an agent who is not successful at selling a property can suffer additional negative consequences for their reputation. I will model this reputation cost as an increasing function of experience $r(e)$. It can be interpreted as both or either a loss of reputation or an opportunity cost - each failed sale precludes the agent to work on another sale. In either interpretation, it is reasonable to assume that the cost is higher for experienced agents. I will verify this model prediction in the data.

[16] This is true as long as $\nu(e)\frac{v(1-\psi-3\%)-\beta V_S}{c_b(1-\psi-3\%)} > e \approx 2.71$. Note that the numerator is the match surplus, $\nu(e)$ is in the 1-3 range. The buyer entry cost c_b is an order of magnitude smaller than the match surplus.

An agent solves the following optimization problem to maximize their utility:

$$\max_{\theta,p} \quad \mu(\theta,e)\psi p - (1-\mu(\theta,e))r(e) + (1-\mu(\theta,e))\beta\alpha(e)V_A(e) \tag{12}$$
$$s.t.$$
$$p = v - \frac{c_b\theta}{1-e^{-\nu(e)\theta}}$$

The continuation value of being an agent of experience e is defined as a contraction mapping:

$$V_A(e) = \mu(\hat{\theta}^a_e, e)\psi\hat{p}^a_e + (1-\mu(\hat{\theta}^a_e, e))(-r(e) + \beta V_A(e)),$$

where $\hat{\theta}^a_e$ and $\hat{p}^a_e$ are the solution to the agent's maximization problem. This solution mirrors the solution to the seller's problem, with $-r(e) + \beta\alpha V_A(e)$ instead of βV_S.

$$\hat{\theta}^a_e = \frac{1}{\nu(e)}\log\left(\frac{v\nu(e)}{c_b} + \frac{(r(e)-\beta\alpha(e)V_A(e))\nu(e)}{c_b\psi}\right) \tag{13}$$

$$\hat{p}^a_e = v - \frac{c_b\log\left(\frac{v\nu(e)}{c_b} + \frac{(r(e)-\beta\alpha V_A(e))\nu(e)}{c_b\psi}\right)}{\nu(e) + \frac{c_b\psi}{v\psi + r(e) - \beta\alpha(e)V_A(e)}} \tag{14}$$

Note that the optimal $\hat{\theta}^a_e$ is higher for the agent than it is for the seller. This means that the agent will choose to list at a lower price than the seller prefers at any level of experience. However, since $r(e)$ is increasing in e, there is an upward pressure on the difference between the two prices as the experience increases.

While $\frac{d\hat{p}^a_e}{d\hat{\theta}^a_e} = \frac{d\hat{p}^s_e}{d\hat{\theta}^s_e}$ remains the same, the $\frac{d\hat{\theta}^a_e}{de}$ is larger and can reverse the sign of the total derivative. If reputation concerns are very important to experienced agents, then their optimal pricing strategy will have a strong negative effect on the optimal price. This could potentially lead to experienced agents setting *lower* prices than inexperienced agents.

The expected welfare of sellers matched with an experience e listing agent who chooses a list price for them is given by the contraction mapping:

$$V^S_{agent_p}(e) = \mu(\hat{\theta}^a_e, e)(1-\psi-3\%)\hat{p}^a_e + (1-\mu(\hat{\theta}^a_e, e))\beta V^S_{agent_p}(e)$$

The extent of the principal agent problem can be measured as the difference between $V^S_{agent_p}(e)$

and $V^S_{seller_p}(e)$ for each experience level. Note that it is unclear whether $V^S_{agent_p}(e)$ is increasing in e. On the one hand, more experienced agents have access to better technology which increases the chance of sale for the seller. On the other hand, more experienced agents may set particularly sub-optimal prices, because their reputation cost of a failed listing may be particularly high.

3 Data

I use CoreLogic's Multiple Listing Service (henceforth MLS) dataset to formally estimate the relationship between agent experience and listing outcomes and use them for model calibration. This part of the analysis closely follows the empirical section in Gilbukh and Goldsmith-Pinkham (2024) using a newer version of the dataset that extends the sample for 8 more years.

MLSs are regional databases that real estate agents use to advertise their listings and access data on all current for sale properties for their buyer clients. CoreLogic provides data for 142 such databases. I choose 38 of them that have consistent coverage from 2000 to 2020. Each MLS contains the universe of listings that have been listed with the help of a real estate agent. For each listing, I have information on detailed property characteristics (e.g. number of bedrooms, bathrooms, exact address, etc.) as well as listing characteristics (e.g. dates associated with listing and sale, list and sale prices). Importantly, I have a unique identifier for the listing agent and, if the sale takes place, a buyer agent. This allows me to see historical activity of real estate agents and define their experience.

3.1 Experience Measure

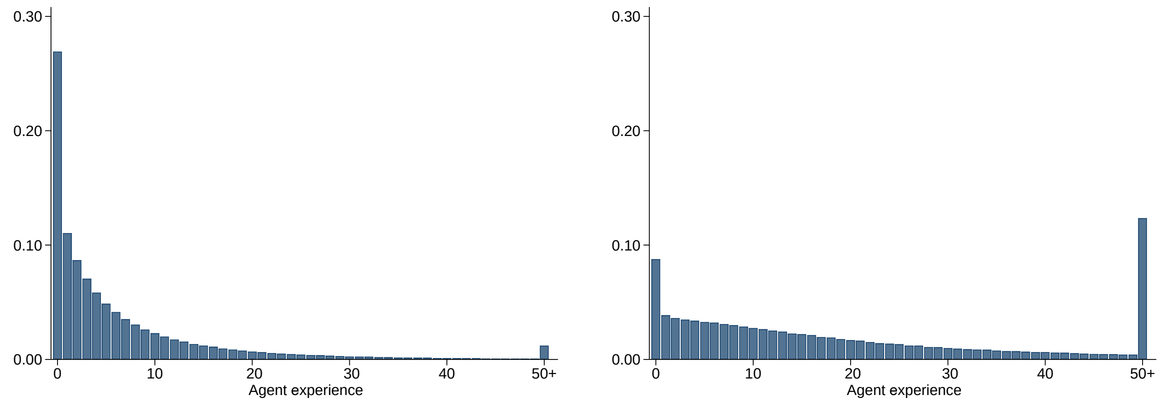
I follow Gilbukh and Goldsmith-Pinkham (2024) and define the agent experience in a particular calendar year as the number of clients they had in the previous calendar year. That is, for each agent, I sum up the number of listings listed in the previous year where they were the primary listing agent and the number of transactions that closed in the previous year where they were the buying agent. Although there are many different ways to define experience, this definition allows for a consistent measure across the sample years that allows me to use the most of the data.[17]

Figure 1(A) plots the distribution of agent experience in the sample. Many listing real estate agents do not have any experience, meaning that no activity was observed in the previous year for these agents. 1(B) plots the distribution weighted by listings that an agent works with. Low-experienced agents have a harder time finding clients, while more experienced agents tend to have more listings.

[17] See Appendix A in Gilbukh and Goldsmith-Pinkham (2024) for comparison of different measures of experience.

Figure 1: Distribution of agent experience

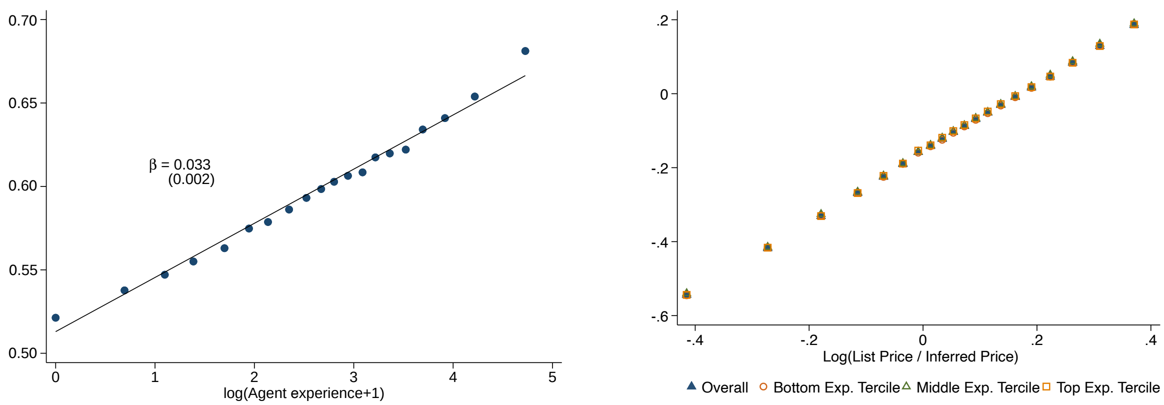
(A) Agent-weighted (B) Listing-weighted



Note: This figure plots the distribution of agent experience. Panel A plots the distribution of experience at the agent-year level. Panel B plots the distribution of experience at the agent-year level, weighted by the number of listings that an agent participated in that year. Agent experience is defined as the number of clients an agent worked with in the previous calendar year.

Figure 2: Agent Experience, Sale Probability, and List Price

(A) Sale Probability (B) List Price (Log)



Note: Both figures are binscatter plots of listing outcome against log of agent experience.

3.2 Experience, Sale Probability, and Prices

I next examine the relationship between listing agent experience and two listing outcomes: sale probability and list prices.[18]. Figures 2(A) and 2(B) shows the binscatter plot of the two outcomes against the log of agent experience controlling for list date and property characteristics (number of bathrooms, bedrooms, square feet).

The relationship between the log of experience and the outcomes is very close to linear. To for-

[18] In the model list price is binding, so is equal to the sale price. In practice, the sale prices can differ from the list price due to bidding wars, negotations, and credits given to buyers post-inspection.

mally estimate it, I run the following regression:

$$y_{i,t} = \alpha_{i,t} + \log(1 + \exp_{i,t}) + \delta W_{i,t} + \epsilon_{i,t}, \tag{15}$$

where $y_{i,t}$ is the outcome for listing i in time t, experience$_{i,t}$ is the experience of the listing agent for listing i in time t, $W_{i,t}$ is a vector of property-specific controls (square footage, the number of bedrooms, and the number of bathrooms), and $\alpha_{i,t}$ denotes month and zip code fixed effects. When the outcome is sale probability or list price, this is a ZIP-code-by-listing-year-month fixed effect, while for sale price outcome this is a ZIP-code-by-closing-year-month effect.

While this OLS is suggestive, it is subject to selection concerns. For example, experienced agents might be working with particularly easy-to-sell properties or easy-to-work with clients. To account for this potential selection bias, I follow the quasi-instrumental strategy outlined in Abaluck et al. (2021).[19] This strategy relies on the fact that sellers often choose a listing agent who they have worked with before. That is, the agent who represented them as a buyer when they purchased the home. However, that agent might not be available for work if they have exited the market. In that case, the seller hires an average listing agent.

To illustrate this strategy, consider seller A and seller B who both bought similar homes in the same location at the same time with buyer agents of the same experience. Suppose that after some time, both homeowners decide to sell the house at the same time. Seller A's agent is no longer available in the market, while seller B's agent is still active. Then seller A is more likely to end up with an "average" agent, while seller B is likely to go back to their experienced buyer agent. As a result, A gets a negative shock to experience relative to B.

For this strategy to work, I need three conditions to be true: 1) inertia - sellers are indeed more likely to work with their buyer agent than they are with a random agent, 2) reversion to mean - when their agent is unavailable, the seller agent's pick looks more like the average agent in the market than their previous agent, and 3) random exit shock - buyer agent subsequent exit is not associated with the home's or the client's unobserved characteristics. I test 1) and 2) in the first stage regression illustrated in Figure 3(A). On the x-axis I plot the experience of the agent who assisted the seller in purchasing the home they are trying to sell. On the y-axis is the experience of the selling agent that seller has hired. The sample is split into seller's whose buyer agent is active in the market, and the sample

[19] The original paper studies the impact of insurance on mortality outcomes. Gilbukh and Goldsmith-Pinkham (2024) is the first paper to adopt this strategy in relation to real estate agents.

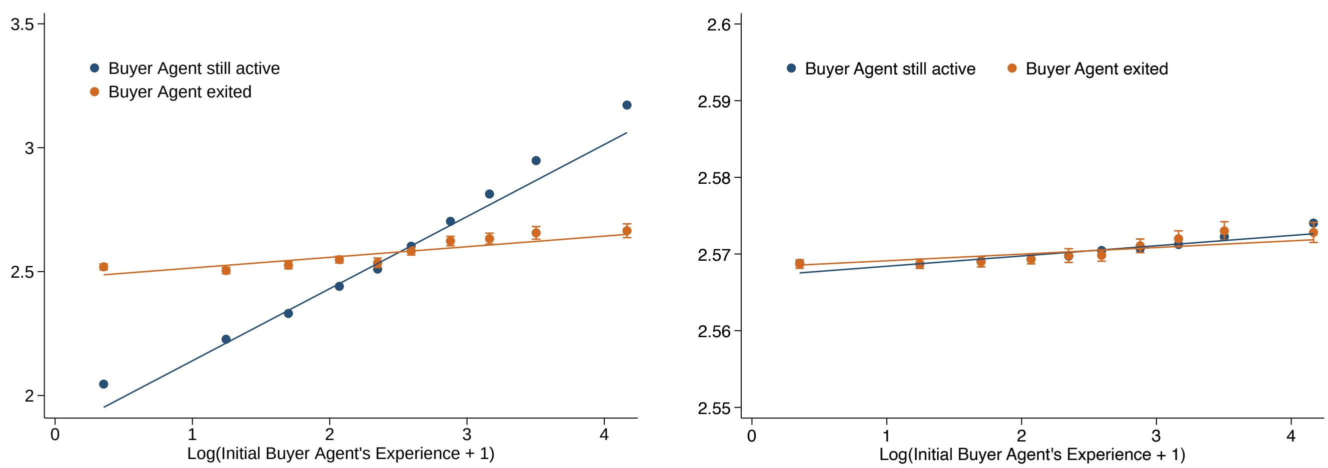
whose buyer agent has exited by the time of the subsequent listing. There is a clear difference in slope between these two lines. For seller's whose buyer agent is available, that buyer agent experience is predictive of their seller agent experience (this is because it is likely to be the same agent, hence, inertia). However, in the sample where the buyer agent has exited, the relationship is much weaker (reversion to the mean). The third condition is also plausibly true. It is unlikely that agent exits are connected with their previously successful buyer agent transaction.

Although it is fundamentally not possible to prove that the exclusion restriction holds, I test for whether any of the observable characteristics of the home are predictive of the relationship between listing agent and previous buyer agent experience. In Figure 3(B) I estimate coefficients on a linear model of listing agent experience as a function of home characteristics. I then predict agent experience for all listings in the data. Finally, I plot the previous buyer agent experience against the predicted seller agent experience in both samples where the buyer agent is and is not active. I find no difference in the two samples.

Figure 3: Instrumental variable strategy with buyer's agent experience & exit

(A) First stage: effect on listing agent experience

(B) Exclusion restriction: effect on predictable agent experience



Note: This figure illustrates the validity of the quasi-experimental design. In Panel A, I plot listing agent experience against the most recent experience of a buyer agent who worked with the seller in the original purchase of the property. In Panel B I do a similar analysis where instead of the actual listing agent experience I plot the predicted experience based on observable housing characteristics. Data in both panels are split by whether the original buying agent is still active at the time of the listing.

Tables 1, 2, and 3 present the results. In column one, I show the outcomes of the OLS regression with month by zip code fixed effects only. In column 2 I add controls for housing characteristics. Column 3 presents results from the IV regression described earlier in the section, and column 4 repeats the regression from column 2 using the IV sample. My preferred specification is the IV regression in column 3[20] I find that increasing the listing agent's experience by one log-point results in 3.1 percentage point (pp) increase in the probability of sale but has no consequential impact on either the list price or the sale price. In my sample, there is a 3 log point difference between agents in a 5th and 95th percentiles of experience. Thus, I conclude that experienced agents are on average 9.3 pp more likely to sell their listings.

Table 1: Effect of experience on probability of sale in 365 days

	(1)	(2)	(3)	(4)
Log(Exp + 1)	0.0232***	0.0211***	0.0313***	0.0262***
	(0.0020)	(0.0024)	(0.0057)	(0.0020)
Month X Zip FE	Yes	Yes	No	No
House Char.	No	Yes	Yes	Yes
Zip X PurchaseYr X ListYr	No	No	Yes	Yes
Direct IV Effects			Yes	Yes
Estimation Procedure	OLS	OLS	IV	OLS
N	3302326	2683103	1642016	1642016

Note: This table reports estimates of the effect of listing agent's experience (using the $\log(1 + \text{experience})$) on a listings' probability of sale in 365 days. All four columns use different versions of the specifications. All columns include ZIP-code-by-listing-year-month fixed effects, and columns 2-4 add controls for house characteristics. Columns 4 and 5 include purchase-year-by-listing-year-by-ZIP-code fixed effects. Column 3 shows results from the IV estimation while column 4 repeats the specification in column 3 using the IV sample. Standard errors are clustered at the MLS level. *$p < .1$; **$p < .05$; ***$p < .01$.

Table 2: Effect of experience on log of list price

	(1)	(2)	(3)	(4)
Log(Exp + 1)	-0.0099*	-0.0000***	-0.0000	-0.0000***
	(0.0055)	(0.0000)	(0.0000)	(0.0000)
Month X Zip FE	Yes	Yes	No	No
House Char.	No	Yes	Yes	Yes
Zip X PurchaseYr X ListYr	No	No	Yes	Yes
Direct IV Effects			Yes	Yes
Estimation Procedure	OLS	OLS	IV	OLS
N	3302326	2683103	1642016	1642016

Note: This table reports estimates of the effect of listing agent's experience (using the $\log(1 + \text{experience})$) on the list price of a listing. All four columns use different versions of the specifications. All columns include ZIP-code-by-listing-year-month fixed effects, and columns 2-4 add controls for house characteristics. Columns 4 and 5 include purchase-year-by-listing-year-by-ZIP-code fixed effects. Column 3 shows results from the IV estimation while column 4 repeats the specification in column 3 using the IV sample. Standard errors are clustered at the MLS level. *$p < .1$; **$p < .05$; ***$p < .01$.

[20]Though the results are closely aligned with the OLS analysis.

Table 3: Effect of experience on log of sale price

	(1)	(2)	(3)	(4)
Log(Exp + 1)	-0.0073*	0.0017**	0.0021	0.0012**
	(0.0043)	(0.0008)	(0.0026)	(0.0006)
Month X Zip FE	Yes	Yes	No	No
House Char.	No	Yes	Yes	Yes
Zip X PurchaseYr X ListYr	No	No	Yes	Yes
Direct IV Effects			Yes	Yes
Estimation Procedure	OLS	OLS	IV	OLS
N	2147020	1754397	1110599	1110599

Note: This table reports estimates of the effect of listing agent's experience (using the $\log(1 + \text{experience})$) on the close price of a listing. The estimation is on a sample of listings that eventually sold. All four columns use different versions of the specifications. All columns include ZIP-code-by-listing-year-month fixed effects, and columns 2-4 add controls for house characteristics. Columns 4 and 5 include purchase-year-by-listing-year-by-ZIP-code fixed effects. Column 3 shows results from the IV estimation while column 4 repeats the specification in column 3 using the IV sample. Standard errors are clustered at the MLS level. *$p < .1$; **$p < .05$; ***$p < .01$.

4 Model Calibration

The model sets up a trade-off for sellers working with agents of different experience. On the one hand, more experienced agents have access to better technology that allows for a more likely match with a buyer. On the other hand, experienced agents have an incentive to set lower list prices in order to avoid the no sale outcome. To see which of the two forces is stronger, I calibrate the model to fit the data.

I set the discount factor β to .9, and the seller commission at $\psi = 3\%$.[21] Additionally, c_b is not separately identified from v as they enter together in the buyer free entry condition, so I set c_b to \$2000. From the data, I can also directly compute the probability that a seller relists with the same agent, conditional on a no-sale outcome $\alpha(e)$. Finally, I compute the average distribution of experience of real estate agents in order to compute seller's ex-ante expectation of putting the house on the market.

I assume that the reputation cost function takes the following form: $r(e) = r_1 log(e)$. That is, I assume that agents with no experience do not suffer any reputation costs from not selling their listings and work to only maximize their income from the transaction. This functional form allows for minimal or no differences between experienced and unexperienced agents. Similarly, I impose ν to have the following functional form: $\nu(e) = \nu_1 e^{\nu_2}$, where experience advantage in the search market is allowed to flexibly vary by experience level.

[21] While the industry standard might shift post the NAR lawsuit of 2024, my dataset excludes the most recent time period and evidence suggests that the commissions are fairly standard during my data sample. See Gilbukh and Goldsmith-Pinkham (2024), Barwick and Wong (2019)

Finally, I assume that seller continuation value is the expected value of listing their home for sale in the following period net of the interest rate they pay on the value of the house for another year. That is, if the seller is unable to sell their house, then they do not have access to the proceeds from sale to invest in their next home and have to borrow that amount from the bank at rate i. The expectation is taken over the listings-weighted distribution of real estate agent experience that they might work with. In other words,

$$V^S = \sum_e pr(e) V^S_{agent_p}(e) - \sum_e i\hat{p}^a_e \tag{16}$$

where $pr(e)$ is the probability of listing with an agent of experience e. I solve for V^S when solving for model equilibrium using the distribution of experience in Figure 1(B).

I set up a grid for parameters ν_1, ν_2, r_1, v to match the average sale probability and the average price, as well as prices and sale probabilities for each experience level.

Table 4: Calibration

Parameter	Value	Identifying Moments
$r(e) = r_1 \times log(e)$		
r_1	\$530	Relative prices by agent experience
$\nu(e) = \nu_1 \times e^{\nu_2}$		
ν_1	0.012	Base sale Probability
ν_2	0.032	Relative sale probability by state
v	485000	price level, jointly identified with c_b
c_b	2000	normalized
β	0.9	-
r_1	0.07	-
$\alpha(e)$	0.3+0.05*log(e)	probability of relist with the same agent
$V^S = \sum_e pr(e)$		distribution of agent experience

Note: This table shows the calibrated values of model parameters and the identifying moments in the data.

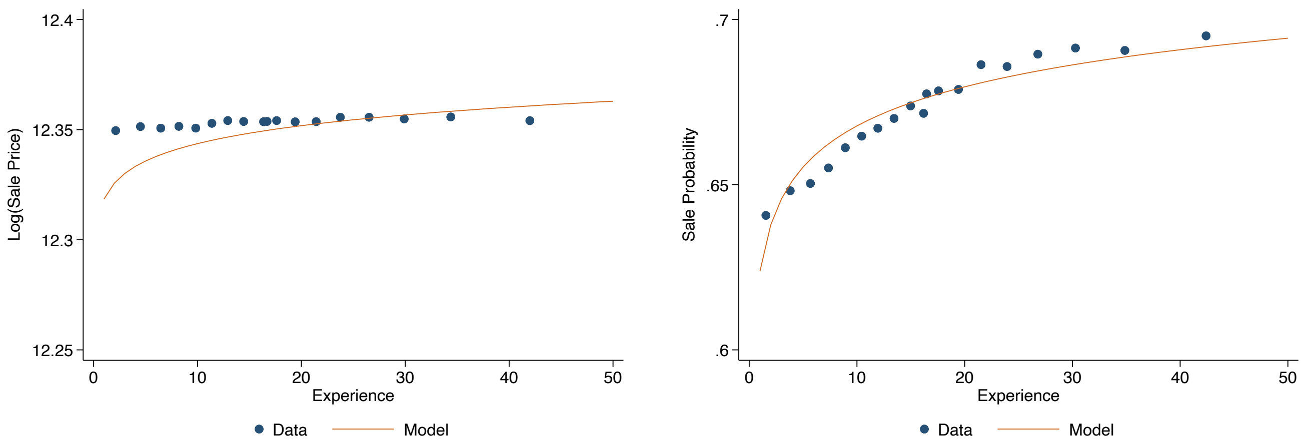
5 Results

5.1 Model Fit

Recall that the parameters were identified by matching sale probability and prices by experience levels. Figure 4 plots the model fit for these variables. The model matches the sale probability well, but does not deliver a completely flat sale price observed in the data. In other words, inexperienced agents in the model set slightly lower prices than experienced agents. This is because the smoothing of the price function is based on a single parameter r_1, taking the functional form of the reputation cost function as given. However, the difference in pricing outcomes between the model and the data is relatively small.

Figure 4: Model Data Match

(A) Log (Sale Price) (B) Sale Probability



Note: This figure illustrates the model fit with the data. Panel A plots the logarithm of a sale price against agent experience. Panel B plot the sale probability of listings by agent experience.

5.2 The Extent of the Principal Agent Problem

Figure 7 shows results from the calibrated model. 5(A) plots the optimal list prices that an agent and a seller would choose for a given agent experience level. Two observations are worth pointing out. First, at any experience level, a seller would like to set a higher price than the agent. This is because a seller is more willing to risk the no-sale outcome than an agent. While a seller can try to sell the property again in the next period, an agent is likely to loose a client and the commission if the sale does not occur within their contract duration. A second observation is that sellers optimally set higher prices as agent experience increases. This is because higher experience agents have access to a better matching technology and sellers want to take advantage of that by raising prices. However, experienced agents also have higher costs in the case of a failed transaction. As a result, experienced agents do not increase their optimal prices, and in fact keep them flat (this is calibrated to the flat relationship between prices and experiences in the data). Next, I plot the sale probability that corresponds to the optimal prices in Figure 5(B). This plot mirrors the previous one. The seller's optimal sale probability is lower than that implicitly chosen by the agent through price. This is because agents would like to avoid the no-sale outcome. In addition, the gradient for the seller-optimal sale probability is higher than the agent-optimal sale probability because of the opposing force of increased reputation concern that comes with agent experience.

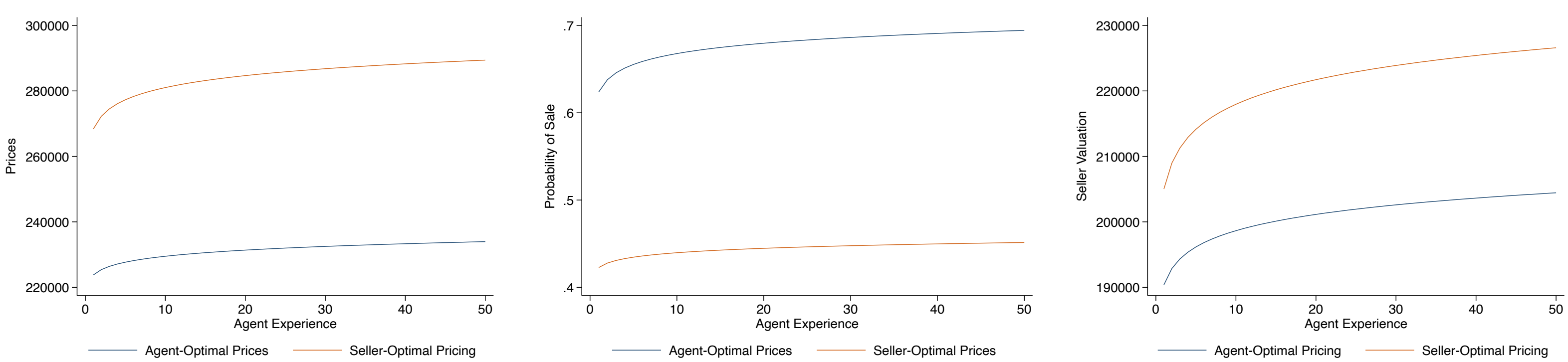
Finally, in 5(C) I plot the welfare of the seller at each level of experience of the agent calculated at the optimal price of the seller (orange line) and the optimal price of the agent (blue line). Since the seller-optimal price is welfare maximizing, the orange line is always above the blue line. However, a seller is always better off with an agent who is more experienced. Two opposing forces are at play defining the shape of the blue line. First, experienced agents have access to better technology and thus are more likely to sell their listings at any price. This force makes experience more valuable to the seller clients. The second force, however, is agent's reputation concern that increases with experience. This makes the principal agent problem more severe for experienced agents as they are setting particularly low prices for their clients in order to avoid the no-sale outcome. This second force could theoretically reverse the positive relationship between seller welfare and their agent's experience. In the calibrated model this is not the case.[22]

Figure 5: Model Results

(A) Seller- and Agent- Optimal Prices

(B) Sale Probability

(C) Overall Seller Welfare



Note: This Figure plots model-generated prices, sale probability, and overall seller welfare from working with agents of different experience. For each plot, the orange line represents the outcome from seller-optimal pricing, while the blue line represents the outcome from agent-optimal prices (i.e. the realized outcomes in the baseline model, since agents set the prices).

6 Alternative Commission Structures

In this section I propose two commission exercises. A unique feature of the model is that agents are allowed to endogenously respond to commission structure by setting optimal list prices. Thus,

[22] This is trivial to see once one realizes that the only difference between agent experience for the seller is the sale probability advantage, since all agents set the same prices.

increased commission rate might actually benefit the seller by better aligning agent's and seller's incentives. The first exercise is similar to the "fair" commission computed in Gilbukh and Goldsmith-Pinkham (2024), but with endogenous prices set by the agents.[23] I assume that inexperienced agents can charge the status quo of 3% and calculate for each experience e, the level of commission that makes sellers indifferent between working with agents of experience e and being represented by someone with no experience at 3%. In the second exercise, I propose a piecewise linear commission structure that accounts for agent heterogeneity and allows sellers to effectively adjust the marginal commission they offer to agents of different experience levels even if experience of their agent is ex-ante unknown. This is because agents with different experience will optimally choose to list properties at the notched prices of the commission schedule.

6.1 "Fair" Commission By Experience

The model has shown that even though the principal-agent problem is more severe among experienced agents, sellers are still better off working with the more experienced agent they can find. What is puzzling in this market is why sellers do not seem to seek experience and why new agents have such a significant portion of the market. Another way to ask the question is why are the commission rates uniform across experience levels? This question is beyond the scope of this paper.[24] However, the model allows me to compute what the competitive commission, i.e. the commission that agents of different experience can charge to make sellers indifferent between them.

If inexperienced agents charge a 3% commission, more experienced agents could increase their commission above 3% to make the client indifferent between working with them and with a new agent. Note that in this equilibrium, seller's valuation is independent of which agent they work for. I can thus solve for seller's continuation value assuming that they always work with an inexperienced agent. I can solve for the contraction mapping W_{flexc}:

$$W_{fair} = \mu(\theta_0^a, p_0^a)p_0^a(1 - 6\%) + (1 - \mu(\theta_0^a, p_0^a))\beta W_{fair}$$

This value is the same as the seller's value of working with an inexperienced agent in the baseline

[23] In Gilbukh and Goldsmith-Pinkham (2024) prices were assumed to remain the same as in the baseline, and inexperienced agents worked for free (competitive commission). Then, for each experience level, a commission was proposed that made sellers indifferent between paying that commission to an agent with a particular experience and getting a free service with an inexperienced agent. Here, I assume that inexperienced agents can charge the status quo of 3%.

[24] It could be, for example, that sellers do not realize the value of agent experience. Alternatively, there are so many agents in the United States that there is inevitably one in everyone's immediate circle of friends and family. Not working with that person can have social consequences, regardless of their experience.

model. For each level of experience, I can then solve for the commission rate ψ_e^{fair} that makes the seller indifferent between working with an agent of experience e and one of experience 0.

$$W_{fair} = \mu(\theta_e^a, p_e^a)p_e^a(1 - \psi_e^{fair} - 3\%) + (1 - \mu(\theta_e^a, p_e^a)p_e^a)\beta W_{fair}$$

Note that p_e^a here is an endogenous variable that depends on the commission level that an agent receives for the transaction. The solution to ψ_e^{fair} for each experience level is thus a fixed point where the combination of ψ_e^{fair} and the corresponding combination of p_e^a and θ_e^a that deliver utility W_{fair} for the seller.

Figure 6(A) displays the resulting commission by experience level. If inexperienced listing agent charges a 3% commission for a successful sale, an agent with high experience level can charge up to 12.57% of the sale proceeds to make the seller indifferent between the two services. Dashed lines in Figures 7(A), 7(B), and 7(C) show prices, sale probability and seller welfare associated with the "fair" commission structure. Note that, by construction, welfare is independent of experience, making sellers indifferent between experience levels of their agent. While experienced agents charge a higher commission (as seen in Figure 6(A)), they deliver higher prices at the expense of only slightly lower sale probability.

6.2 Piecewise Linear Commission Structure

One can think of the "fair" commission exercise as predictive of what commission rates would arise if real estate agent market were more competitive. Experienced agents would be able to dictate a higher price and still be in demand by sellers. This, of course, would only be possible if agent experience was an observable feature of the service. In the second exercise, I consider an alternative commission structure in which the marginal compensation of the listing agent varies with price, but is the same for agents of all experiences. In particular, I consider a piecewise linear commission step function.[25] While the same commission schedule would apply to all agents, those with more experience would find it worthwhile to aim for a higher price and thus higher marginal compensation, while those with little experience would forgo risking a "no-sale" outcome for a higher commission. To allow for most flexibility, I will look for a piecewise linear commission defined by price notches $\{p_1, p_2, ..., p_50\}$ (one for each level of agent experience) and marginal commission amounts $\{\psi_1, \psi_2, ..., \psi_{50}\}$ corresponding to price intervals defined by these notches. The total commission an agent earns for selling the house

[25] A simple example is compensation of 3% up to $100,000, and 10% for the amount of the final price above $100,000. Here, if an agent were to sell the house for $150,000, they would earn 3% * $100,000 + 10% * ($150,000-$100,000) = $8,000.

at price p^* is the area under the commission step function between $p = 0$ and $p = p^*$. Formally, the total commission from the transaction at price p is equal to

$$\psi(p) = \psi_1 \min(p, p_1) + \sum_{e \in \{2,3,...50\}} \psi_e(min(p, p_e) - min(p, p_{e-1})) \tag{17}$$

This more general set-up allows for marginal commissions to be targeted to each experience level, thus allowing the seller to extract the most welfare from working with any agent, even if their experience is ex ante unknown. I next describe the solution method for finding optimal price notches and marginal commissions corresponding to each experience level.

I first assume that $\psi_1 = 3\%$ and p_1 is the price that agent of experience one charges in the baseline model. I then use an iterative procedure for each subsequent experience level $e \in \{2, 3, ..., 50\}$ to find p_e and ψ_e. For each e I perform a grid search over possible commission rates $\psi_e \geq \psi_{e-1}$, solving for corresponding optimal pricing strategies p_e of an agent with experience e (I describe this step in more detail in the next paragraph). I select the commission ψ_e that maximizes the utility of the seller if they were matched with an agent of experience e. The price p_e corresponding to ψ_e is taken to be the next level in the commission step function. Finally, I verify that with the additional notch, it is still optimal for agents with experience less than e to stick to the pricing strategies defined by $\{p_1, p_2, ..., p_{e-1}$ found in the previous steps.[26]

I next describe the optimal pricing strategy of an agent facing a piecewise linear commission structure. Formally, I have to solve for an optimal price within each price segment corresponding to different marginal commissions and then choose among those the price that delivers the most expected profit to the agent. In practice, the optimal price of an agent with experience e will be higher than p_{e-1}, so finding the optimal price on the last segment is sufficient.[27]

For each commission schedule defined by $\{p_1, ..., p_{e-1}\}$ and $\{\psi_1, ..., \psi_e\}$ I then solve the following problem:

[26] In that sense I am using a pareto-improving commission schedule, where the inexperienced agents find it optimal to stay at status quo, while more experienced agents can benefit from a higher commission set optimally by the seller.

[27] Agents trade off probability of sale and the price - trying to get the best commission while considering that no-sale leads to zero income and an additional cost. Our empirical analysis shows that this trade-off is essentially the same for agents of all experience levels at $\psi = 3\%$. However, as marginal commissions increase, experienced agents would find it optimal to increase prices more than inexperienced agents because they would find it less costly to do so. That is because for the same price increase, an experienced agent will have a more likely sale.

$$\max_{\theta,p} \quad \mu(\theta,e)(\bar{\psi}+\psi_e(p-p_{e-1}))+(1-\mu(\theta,e))(\beta V_A(e)-r(e)) \tag{18}$$

$$s.t.$$

$$p = v - \frac{c_b\theta}{1-e^{-\nu(e)\theta}} \tag{19}$$

$$\bar{\psi} = \psi_1 p_1 + \sum_{j\in\{2,\ldots,e-1\}} \psi_j(p_j - p_{j-1})$$

That is, an agent with experience e maximizes their profit by choosing p and the corresponding θ (the trade-off set by buyer sorting). If they sell the home (with probability $\mu(\theta,e)$), they receive a constant commission $\bar{psi}$ for selling above p_{e-1} and the marginal commission ψ_e on every additional dollar above p_{e-1}. With the complementary probability, a no-sale outcome is realized and the agents receive a discounted continuation value $V_A(e)$ and incur a cost $r(e)$. Agents take into account the trade-off between the probability of sale and prices and can compute $\bar{psi}$ from the commission schedule as the area under the marginal commission function of sale price.

The continuation value of the agent is a contraction mapping:

$$V_A(e) = \mu(\theta,e)(\bar{\psi}+\psi_e(p_e-p_{e-1}))+(1-\mu(\theta,e))(\beta V_A(e)-r(e)) \tag{20}$$

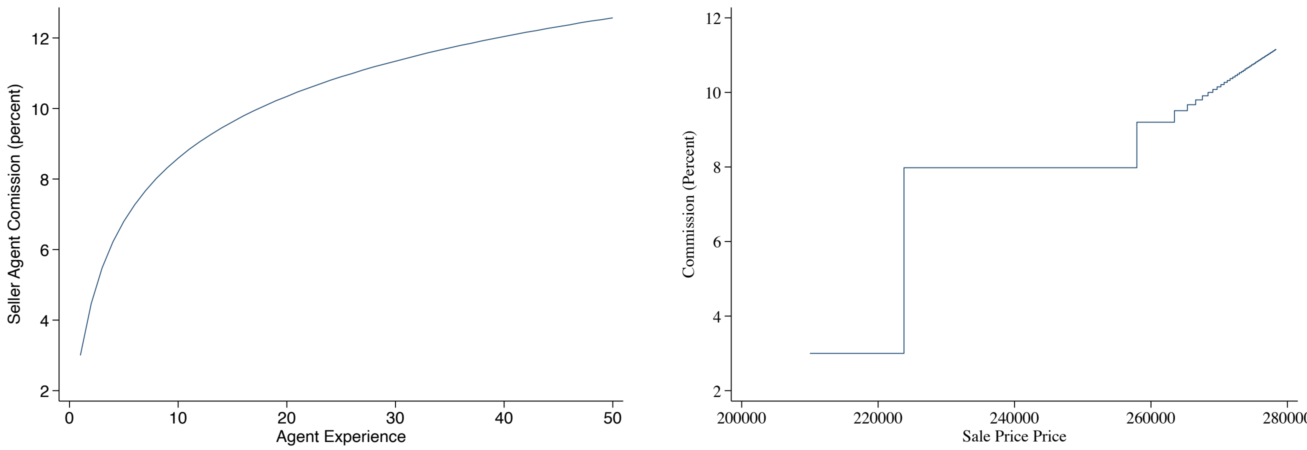
The solution for this problem follows the same steps as one for the original problem with a flat commission schedule. The only difference is the additional constant $\bar{psi}$.

Note that this commission structure is always pareto-improving for the seller relative to the flat 3% compensation. This is because marginal commissions are chosen to maximize the seller's utility. And keeping the commission flat so that $\psi_1 = \psi_2 = ...\psi_{50} = 3\%$ is in the seller's choice set.

Figure 6(B) plots the commission schedule for this exercise. Beginning with a standard 3% compensation for prices below p_1, agents face increasing incentives to try to sell the house for a better price and are compensated as much as 12% for every additional dollar. Dark red lines in Figures 7(A), 7(B), and 7(C) show prices, sale probability and seller welfare associated with the piecewise commission schedule. By construction, working with an agent of experience one leads the same outcome as in the baseline, flat 3% commission case. However, more experienced agents find it worthwhile to increase their prices and take advantage of the increased marginal commission. The resulting pricing is much closer aligned with seller-optimal pricing plotted in orange[28]. The resulting sale probability is lower as higher priced properties are less likely to sell. While in this exercise, sellers have to pay on average higher commissions, their welfare is much improved relative to the baseline plotted in green and is in fact close to welfare obtained by seller-optimal pricing.

Figure 6: Alternative Commission

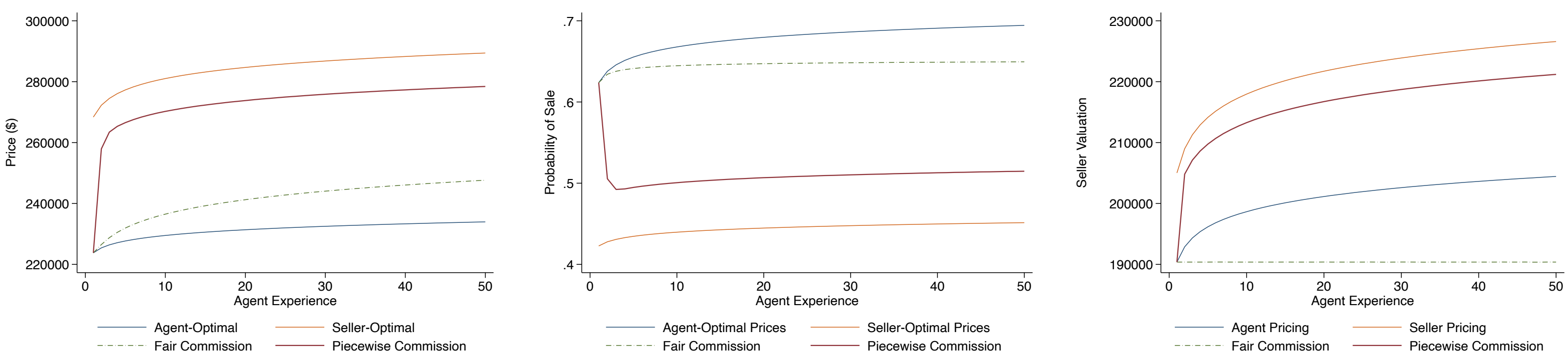
(A) "Fair" Commission (B) Piecewise Linear Commission



Note: This figure plots the outcome of the two commission exercises described in section 6. Panel A plots flat commission rates for each experience level that would make sellers indifferent between all experience options. Panel B plots the optimal piecewise linear commission schedule that would allow marginal compensation vary by the sale price obtained by the agent.

Figure 7: Alternative Commission: Results

(A) Prices (B) Sale Probability (C) Seller Welfare



Note: This figure plots outcomes from the baseline model as in figure 7 together with the outcomes generated by the two commission exercises described in section 6.

[28]Though the seller optimal pricing exercise assumed the commission is constant at 3%.

7 Discussion

One of the strongest assumptions driving the quantitative results of the model is the continuation value of the seller in the case of a failed sale. While it is true that sellers get to keep the house and try the market in the next period, sellers could find it particularly costly to hold on to the property for another year. Some of these costs are accounted for in r, the interest rate on the property value. But there might be additional burdens like maintenance, inability to secure financing for the next property, foregone job opportunities, as well as market risk. Moreover, these can vary from seller to seller as well as by time period. I argue that while the quantitative results might change as a factor of these variables, qualitatively they do not depend on these assumptions. This is because the calibration is done on the agent-optimal pricing that does not depend on seller valuation.

In addition, the model does not allow agents to optimally pick their level of effort, which can affect their pricing strategy. Instead, I assume that agent advantages are based on expertise only. As a result, some interesting compensation structures are outside the scope of the model. For example, if agents were paid a fixed fee upon sale, the model would predict prices to be at the minimum level acceptable to the seller - this lowest price would attract as many buyers as possible to the property and maximize sale probability. Similarly, if agents were compensated per hour, agents would set prices at a particularly high level to charge the seller more money. Only the most experienced agents would have an incentive to sell, in order to avoid paying the reputation costs of a no-sale outcome.

Finally, my model does not consider heterogeneity on either the seller or the buyer side. However, on the seller side, the model can still speak to the heterogeneity affecting preferences for the trade-off between prices and the sale probability. This is most easily illustrated in the "fair" commission exercise where, without the heterogeneity, sellers are indifferent between agents. If for some sellers the no-sale outcome is particularly costly (for example, if they need financing for their future house as soon as possible), then they would prefer to work with experienced agents and pay a higher commission fee. On the other hand, sellers who are in no rush to sell and for whom access to cash is not urgent might prefer to work with an inexperienced agent. They would wait longer for their home to be sold, but would save on commission fees.

8 Conclusion

Real estate agent profession has come under much scrutiny and some argue that the principal agent problem prevents them from providing valuable service to their clients. I build a model to quantify the principal-agent problem for agents of different experience levels and find it is more pronounced in experienced listing agents whose incentives are least aligned with the sellers. This is because experienced agents find failure to close a deal more costly, either because of reputation concerns or simply due to the opportunity cost of their time. Although the principal-agent problem is more severe in top agents, I also show that they more skilled at matching their clients at any price and ultimately deliver a higher welfare to the seller than inexperienced agents. The calibrated model counteracts the two forces and shows that the technology premium is more valuable to the seller than the principal-agent problem is costly. Using the model, I simulate the competitive commission market and show that experienced agents are worth as much as 12% more of the house price to sellers than new agents without experience. To align seller and agent incentives, I propose an alternative compensation strategy of a piecewise-linear commission. I find that with minimal cost for the seller (due to higher marginal commissions), agents will choose to set prices close to the seller-optimal levels and all but erase the welfare loss associated with the principal agent problem.

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